

Week 2 Resources: Machine Learning for Forecasting

The goal of this week is to build a strong intuition for some of the most widely used machine learning models before applying them to time series forecasting. Focus on understanding how these models make predictions rather than the mathematical derivations behind them.

Phase 1: Understanding the Models

1. Linear Regression

Resources

- StatQuest: Linear Regression Clearly Explained
 Linear Regression, Clearly Explained!!!
- Scikit-Learn Documentation
https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LinearRegression.html

Key Takeaways

- Features vs target variable
 - Fitting a model to data
 - Making predictions
 - Understanding coefficients
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2. Decision Trees

Resources

- StatQuest: Decision Trees Clearly Explained
 Decision and Classification Trees, Clearly Explained!!!
- StatQuest: Regression Trees Clearly Explained
 Regression Trees, Clearly Explained!!!
- Scikit-Learn Documentation
<https://scikit-learn.org/stable/modules/generated/sklearn.tree.DecisionTreeRegressor.html>

Key Takeaways

- How trees split data
 - Decision boundaries
 - Overfitting intuition
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3. Random Forests

Resources

- StatQuest: Random Forests Clearly Explained
 StatQuest: Random Forests Part 1 - Building, Using and Evaluating
- Scikit-Learn Documentation
<https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestRegressor.html>

Key Takeaways

- Ensemble learning
 - Bagging
 - Why multiple trees outperform a single tree
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4. XGBoost

Resources

- StatQuest: XGBoost Clearly Explained
 XGBoost Part 1 (of 4): Regression
- Official Documentation
<https://xgboost.readthedocs.io/en/stable/>

Key Takeaways

- Boosting intuition
 - Sequential error correction
 - Why XGBoost performs so well on tabular datasets
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5. Evaluation Metrics

Resource

- Regression Metrics Explained
<https://www.geeksforgeeks.org/machine-learning/regression-metrics/>

Key Takeaways

- MAE (Mean Absolute Error)
 - RMSE (Root Mean Squared Error)
 - Interpreting model performance
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Phase 2: Applying Machine Learning to Time Series

Once you are comfortable with the models above, go through the following Kaggle course (expected time to complete - 4 hours) to understand how machine learning can be applied to forecasting problems.

Kaggle Time Series Course

<https://www.kaggle.com/learn/time-series>

Recommended Lessons:

1. Linear Regression With Time Series
 2. Trend
 3. Seasonality
 4. Time Series as Features
 5. Forecasting With Machine Learning
 6. Hybrid Models
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Highly Recommended Notebook

Time Series as Features

<https://www.kaggle.com/code/ryanholbrook/time-series-as-features>

This notebook demonstrates how feature engineering can transform time series forecasting into a standard machine learning problem using models such as Linear Regression, Random Forests, and XGBoost.
